leaving the house to go to work, and you then look at the weather forecasting app on your phone.

Needless to say, the predictions made by the app need to be accurate and reliable in order for you to base your decisions on. An app that always predicts sunny weather just won't do. Further, you need a sensible way to make decisions from these predictions. Never bringing an umbrella regardless of how likely rainy weather is, is a bad decision-making policy and will get you in trouble when it does rain. On the other hand, always bringing an umbrella, even with 100% sunny weather, is also not a smart decision. You want to *adaptively* decide to bring your umbrella based on the weather forecast.

Adaptively making decisions is what Bayesian optimization is all about, and in order to do it effectively, we need both a good predictive model and a good decision-making policy. Care needs to go into both components of the framework; this is why the two main parts of the book, following this chapter, will be about modeling with GPs and decision making with Bayesian optimization policies, respectively.

1.2.4 Bayesian optimization in action

At this point, you might be wondering if all of this heavy machinery really works, or works better than some simple strategy like random sampling. So let us take a look at a "demo" of Bayesian optimization on a simple function—it will also be a good way for us to move away from the abstract and get more concrete, and tease out what we will be able to do in future chapters.

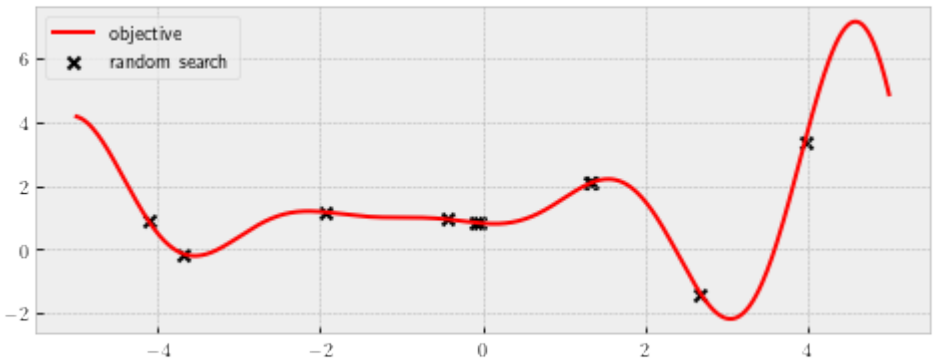


Figure 1.7 The objective function that is to be maximiized, where random search wastes resources on unpromising regions.

Let's say the black-box objective function that we are trying to optimize (specifically in this case, maximiize) is the one-dimensional function in Figure 1.7, defined from -5 to 5. Again, this picture is only for our reference; in black-box optimization, we in fact do not know the shape of the objective. We see that the objective has a couple of local maxima around -5, roughly -2.4, and 1.5, but the global maximum is on the right at approximately 4.3. Let's also assume we are allowed to evaluate the objective function at most 10 times.

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